

AX Auction — Pre-Deployment Simulation Plan

Performance Test Execution Plan

Date: 29 June 2026

Environment: Staging

System: api-gateway.ax-stg.lgq-stg.com

1 How Tests Are Executed

Each test runs **two components simultaneously** to reflect actual production conditions where users perform different activities at the same time.

Main Test

Simulates users performing the full bidding flow:
Login → Browse Catalog → Enter Auction Room → Place Bid → Receive Broadcast

AXOtherOperation (Background)

Simulates other users doing general activities such as browsing the catalog, checking the auction schedule, and viewing notifications — running in parallel to replicate real production load where not all users are actively bidding.

i) Both components run at the same time throughout the entire duration of each test.

2 Server Specification

Component	Current Staging	Upgraded Staging (Prod-Spec)
App Server	2 Core 4 GB RAM	TBD (match production)
Database	4 Core 16 GB RAM	TBD (match production)

3 Test Execution Plan

Tests must be executed in order. Each test builds on the confidence established by the previous one. The upgraded staging config uses higher user counts because the additional server resources are expected to support a higher capacity.

#	Test Type	Current Staging Config	Upgraded Staging Config (Prod-Spec)	Pass Criteria
1	Load Test	120 users, ramp-up 300s, 1 loop	200 users, ramp-up 300s, 1 loop	Error rate 0% Avg response < 200ms
2	Loop Test	60 users, ramp-up 300s, 50 loops (~15 min)	100 users, ramp-up 300s, 50 loops (~15 min)	Error rate 0% Response time consistent per loop
3	Spike Test	Phase 1: 10→120 users Phase 2: 0→120 users sudden spike	Phase 1: 10→200 users Phase 2: 0→200 users sudden spike	Error rate 0% Recovery < 1 minute after spike
4	Endurance Test	60 users, ramp-up 300s, 200 loops (~45 min)	100 users, ramp-up 300s, 200 loops (~45 min)	Error rate < 0.1% Response time stable throughout
5	Stress Test	150 users, ramp-up 60s, 1 loop	200 users, ramp-up 60s, 1 loop	Breaking point ≥ 200 users